DO NOW — RECALL

Lesson 124 · Procedural Programming: Modules and Sta

Name: _____ Class: _____ Date: _____

1. Which paradigm describes a desired result, rather than detailed step-by-step changes?

2. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

3. For the binary value 1010, what is its decimal value?

4. A linked-list node has data and what other essential field?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Declarative programming.

2. Model answer

Sum 0 and carry-out 1.

3. Model answer

10.

4. Model answer

A link/reference to the next node (for a singly linked list).

By the end, you can…

• Statements execute in an order; selection and loops direct flow.

• Reusable named routines accept parameters; functions return values.

• Local variables limit unwanted side effects; global state needs care.

Vocabulary — know these terms

procedure — Named subroutine mainly performing an action.

function — Subroutine that returns a value.

parameter — Named input used by a routine.

scope — Region where an identifier may be accessed.

NEW LEARNING

Read, annotate and explain

Sequence and control

Statements execute in an order; selection and loops direct flow.

Procedure/function

Reusable named routines accept parameters; functions return values.

Scope and state

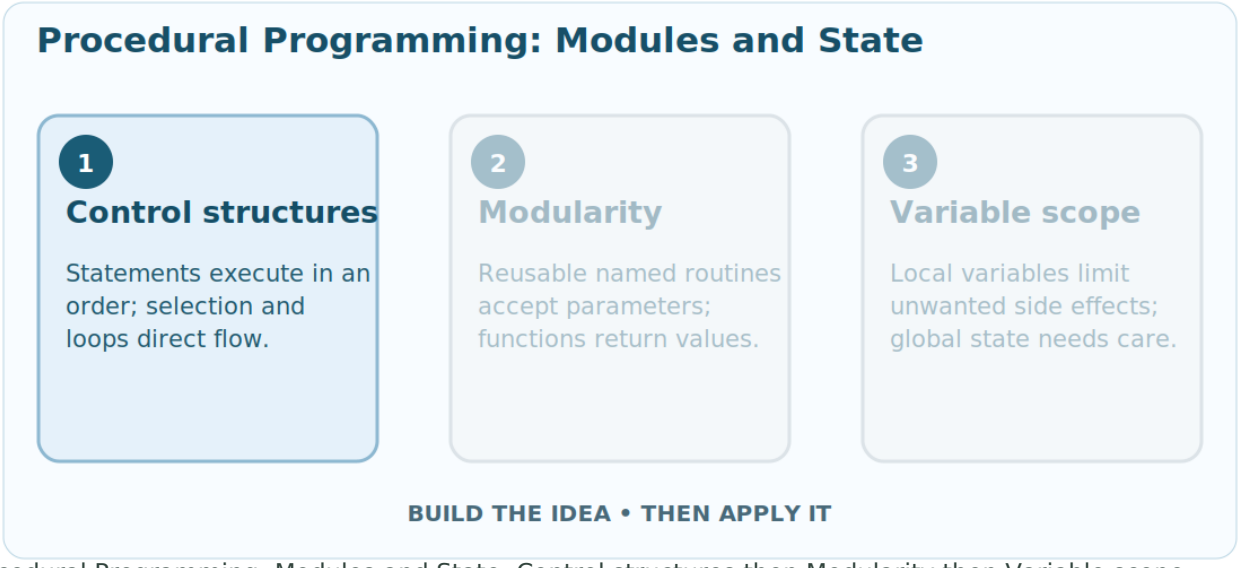
Local variables limit unwanted side effects; global state needs care.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

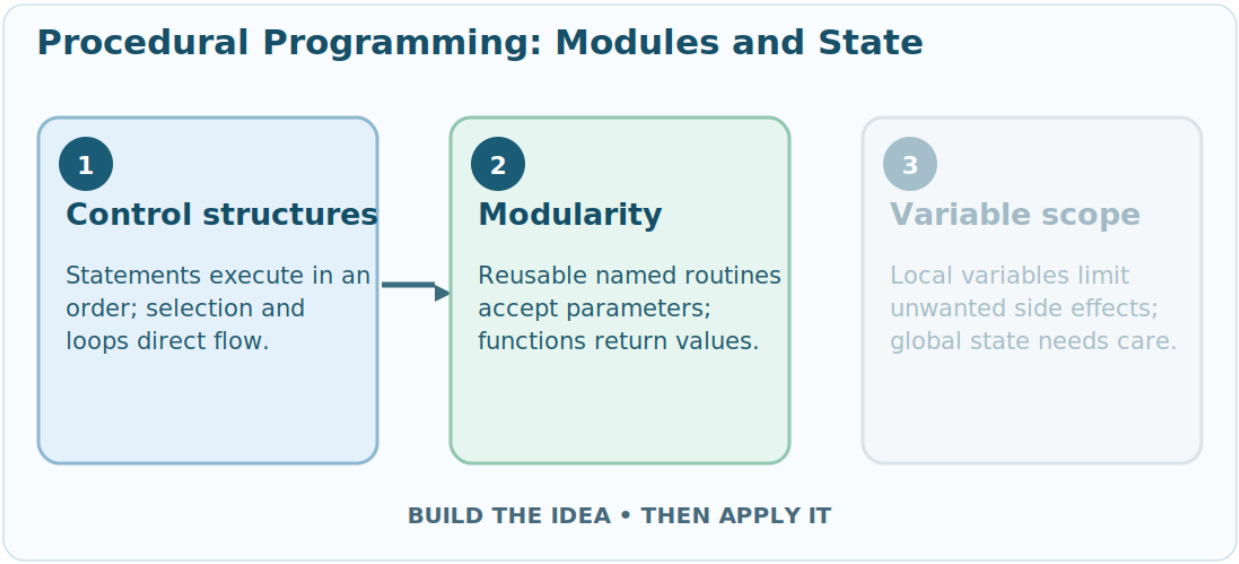
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Procedural Programming: Modules and State: Control structures then Modularity then Variable scope

New Learning

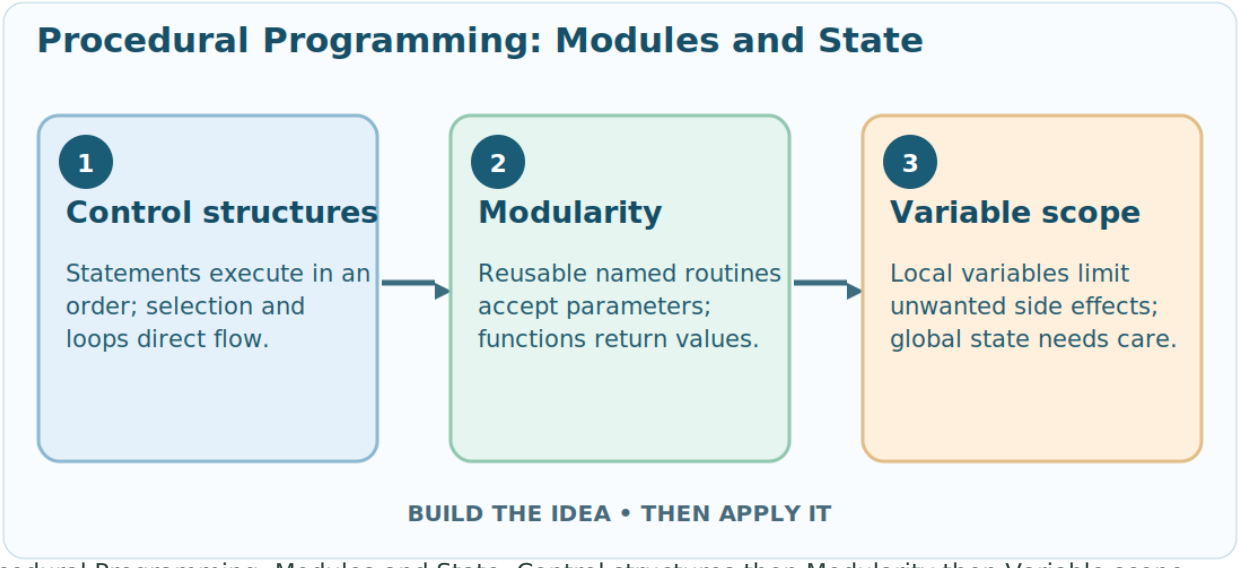


Procedural Programming: Modules and State: Control structures then Modularity then Variable scope

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Procedural Programming: Modules and State: Control structures then Modularity then Variable scope

I DO — WORKED EXAMPLE

Follow the worked method

Trace FUNCTION doublePlusOne(x): result=x*2; RETURN result+1; print(doublePlusOne(4)).

Model answer

Parameter x=4, local result=8, return value 9, printed output 9. The local result does not become a global variable merely because the function runs.

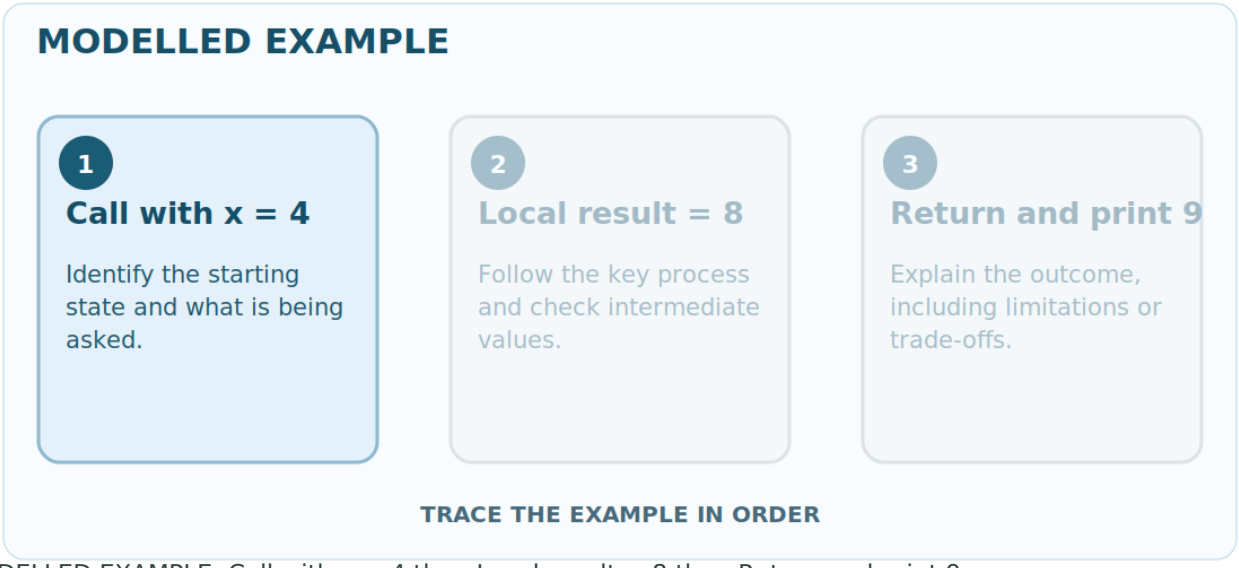
Check: link each claim to the scenario and use precise OCR terminology.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace FUNCTION doublePlusOne(x): result=x*2; RETURN result+1; print(doublePlusOne(4)).

Parameter x=4, local result=8, return value 9, printed output 9. The local result does not become a global variable merely because the function runs.



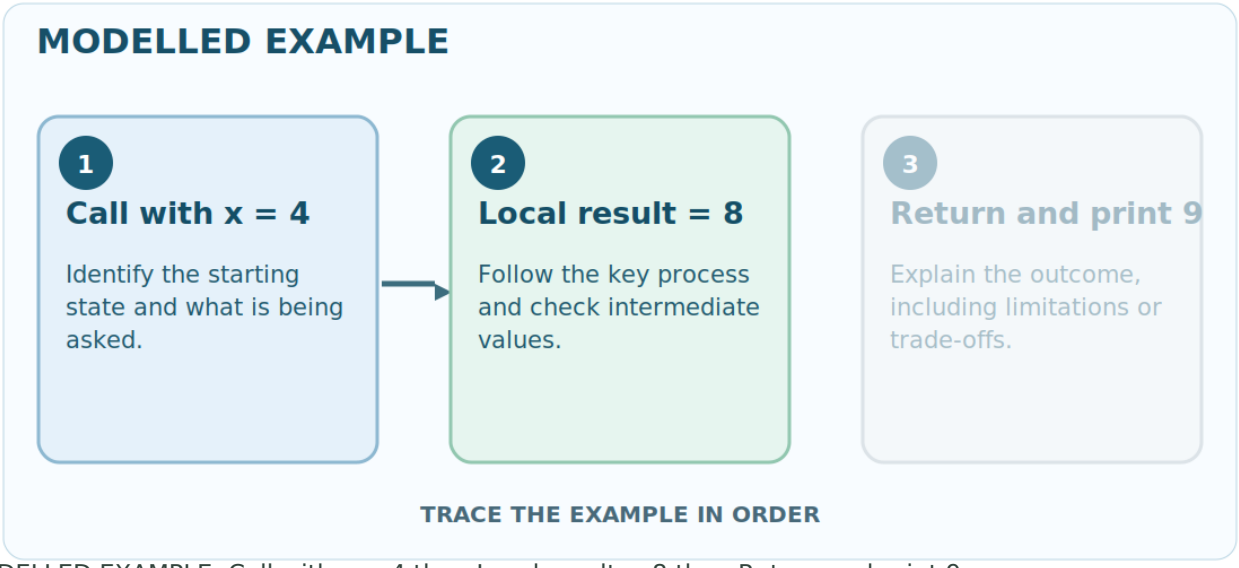
MODELLED EXAMPLE: Call with x = 4 then Local result = 8 then Return and print 9

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace FUNCTION doublePlusOne(x): result=x*2; RETURN result+1; print(doublePlusOne(4)).

Parameter x=4, local result=8, return value 9, printed output 9. The local result does not become a global variable merely because the function runs.



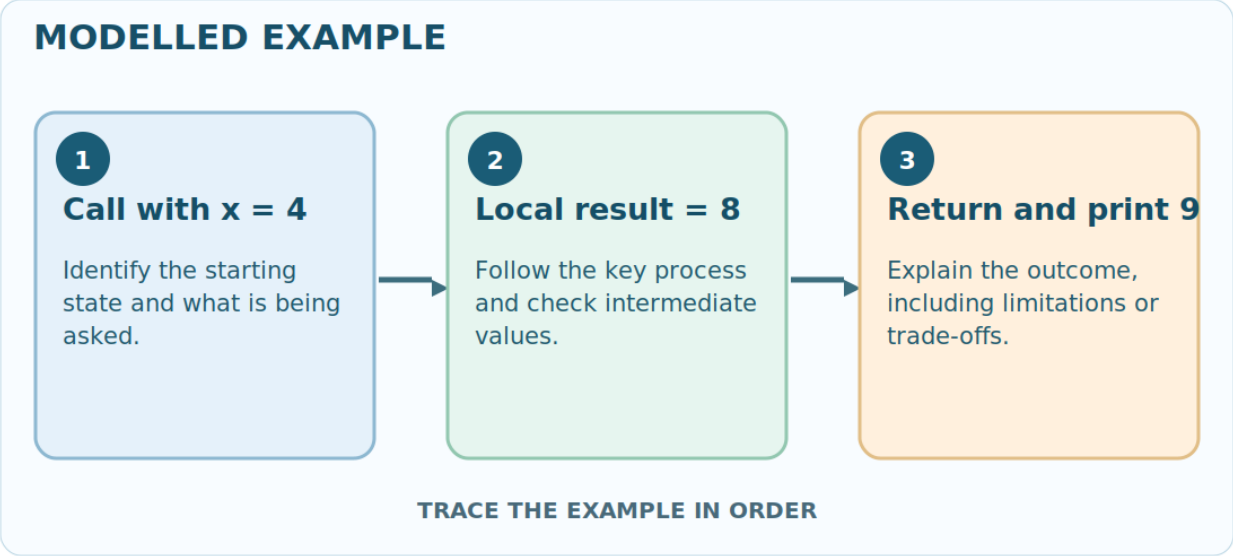
MODELLED EXAMPLE: Call with x = 4 then Local result = 8 then Return and print 9

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace FUNCTION doublePlusOne(x): result=x*2; RETURN result+1; print(doublePlusOne(4)).

Parameter x=4, local result=8, return value 9, printed output 9. The local result does not become a global variable merely because the function runs.



MODELLED EXAMPLE: Call with x = 4 then Local result = 8 then Return and print 9

WE DO — GUIDED PRACTICE

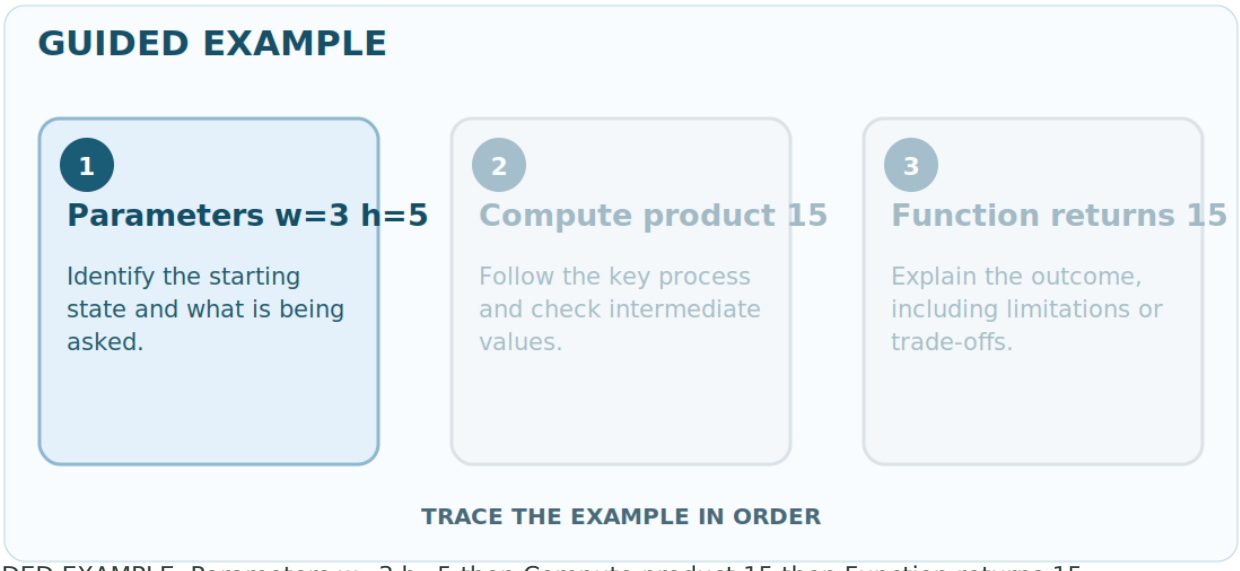
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace FUNCTION area(w,h): RETURN w*h; print(area(3,5)). How would a procedure differ?

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Parameters w=3 h=5 then Compute product 15 then Function returns 15

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the benefits of using functions/procedures rather than one long main routine. [6]

Question 2

Trace FUNCTION f(n): a=n+2; RETURN a*3; OUTPUT f(2). [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain local/global scope and give a bug caused by modifying shared global state. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A local variable declared inside a function is automatically accessible everywhere in the program.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

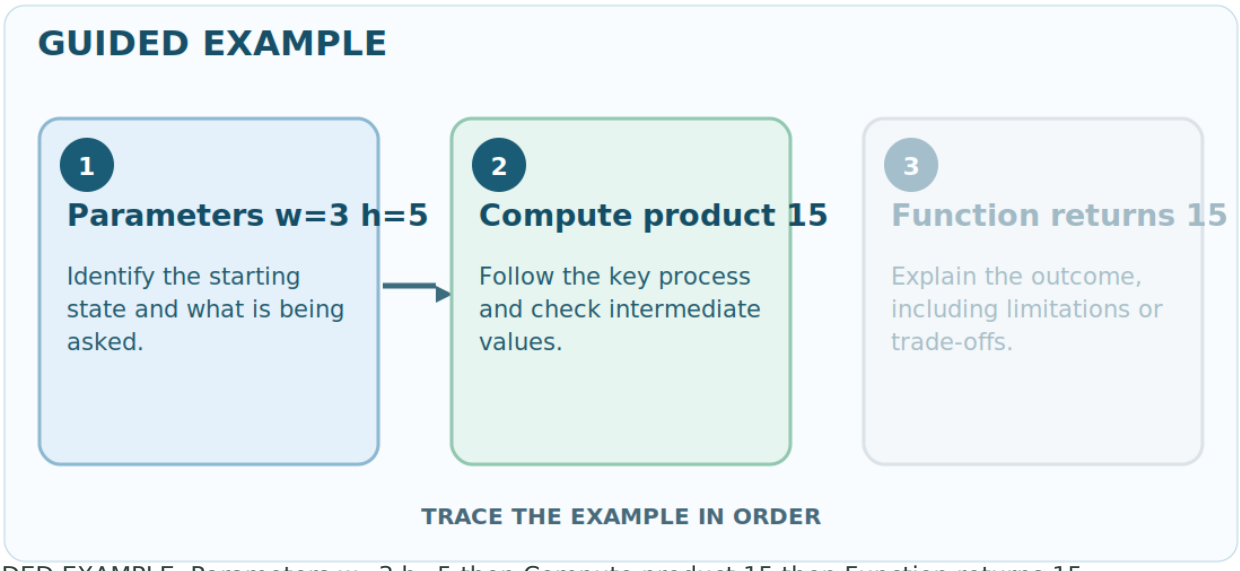
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace FUNCTION area(w,h): RETURN w*h; print(area(3,5)). How would a procedure differ?

Identify the inputs, relevant facts and exact output or judgement required.



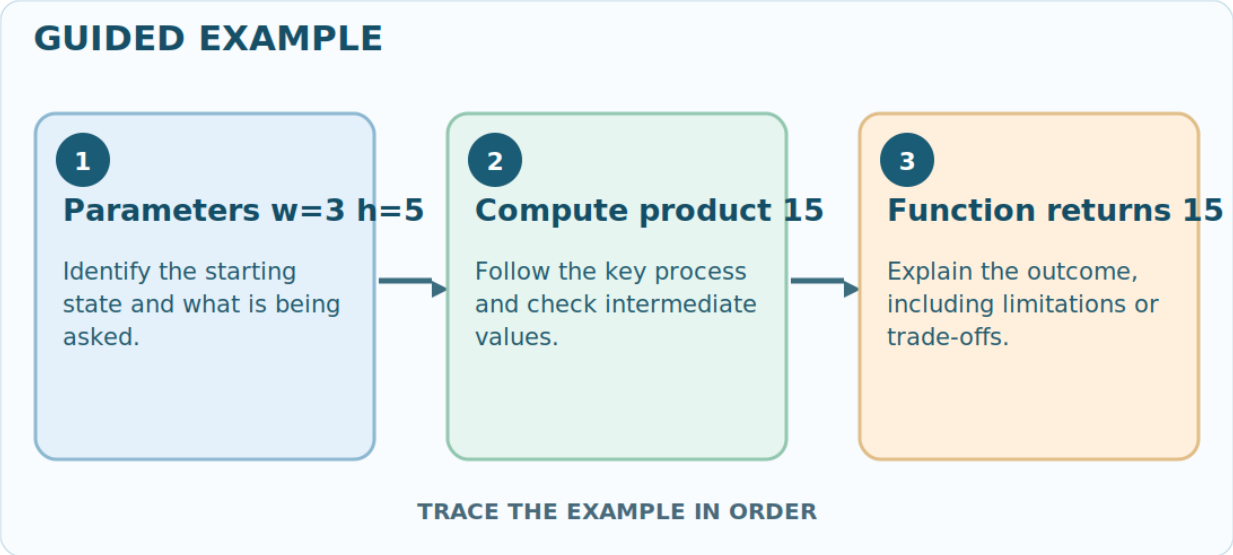
GUIDED EXAMPLE: Parameters w=3 h=5 then Compute product 15 then Function returns 15

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace FUNCTION area(w,h): RETURN w*h; print(area(3,5)). How would a procedure differ?

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Parameters w=3 h=5 then Compute product 15 then Function returns 15

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Arguments bind w=3 and h=5; return 15; print shows 15. A procedure is intended primarily to perform an action rather than return a result (conventions vary by language).

IMPROVE: Explain one correction to your own reasoning.